Non-Intrusive Available Bandwidth Estimation using One-Way Delay Measurements

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Introduction

Estimating available bandwidth (available-bw) has become a crucial task for application owners, network operators and end users. Application managers primarily need bandwidth estimation in order to utilize the available resources at maximum efficiency and provide the best Quality of Service (QoS) to the client. Network operators need this same estimate to plan capacity upgrades for their infrastructure. End users need to know their available-bw to verify that their Internet Service Provider (ISP) is fulfilling their contract of promised bandwidth.

Generally, the available-bw is defined as the maximum rate that a path can provide to a flow, without reducing the rate of the cross-traffic in the path [1]. In the last 30 years, many active estimation techniques and software tools have been proposed. These tools either implicitly or explicitly are built on Link Congestion Effect (LCE): if source host sends probe packets at a rate x that is higher than available-bw, there must be at least one link that forwards the probe packets at a rate lower than x [2].

We may assume that the path a packet takes from end hosts is stable for some period of time, as observed by Paxson et al in [3]. Moreover, it was found that path throughput remains constant for extended periods of time by Zhang et al in [4]. Thus, we can expect experiments to provide consistent measurements for some extended periods of time.

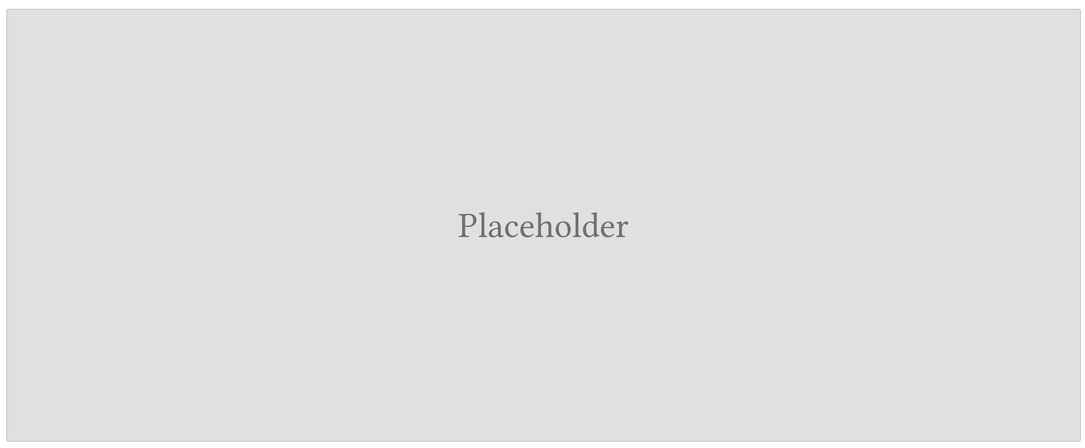

Figure 1:

Background

A network path $\mathcal{P}$ is a sequence of n store-and-forward links that trasfer packets from the source host $\mathcal{R}_0$ to a destination host $\mathcal{R}_n$ through a set of intermediate routers $\mathcal{R}_1, ..., \mathcal{R}_{n-1}$. Link L_i is the data connection from $\mathcal{R}_i$ to $\mathcal{R}_{i+1}$. At any time, L_i is either idle or forwarding packets at its capacity C_i. The end-to-end capacity C is defined as

$$C \equiv \min_{i=0,...,n} C_i$$

and it is the maximum rate that the path can provide to a flow, when no traffic is present in $\mathcal{P}$. The available-bw of this link is then defined as the unused bandwidth over a time interval τ:

$$A_i(t, t+\tau) := \frac{1}{\tau} \int_t^{t+\tau} (C_i - \lambda_i(t))dt,$$

where λ_i is the rate of cross-traffic in L_i. In particular, the *available-bw* that we estimate is the path available-bw. That is, the minimum link available-bw of the target path:

$$A(t, t+\tau) := \min_{i=0,...,n} A_i(t, t+\tau)$$

The link that mantains the minimum link available-bw is named the bottleneck link.

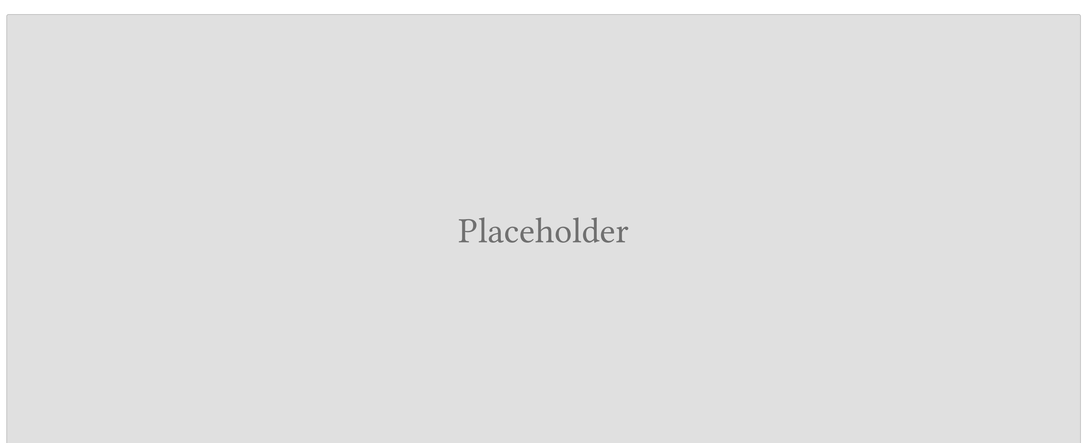

Figure 2:

Related Work

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Figure 3:

Current Findings and Results

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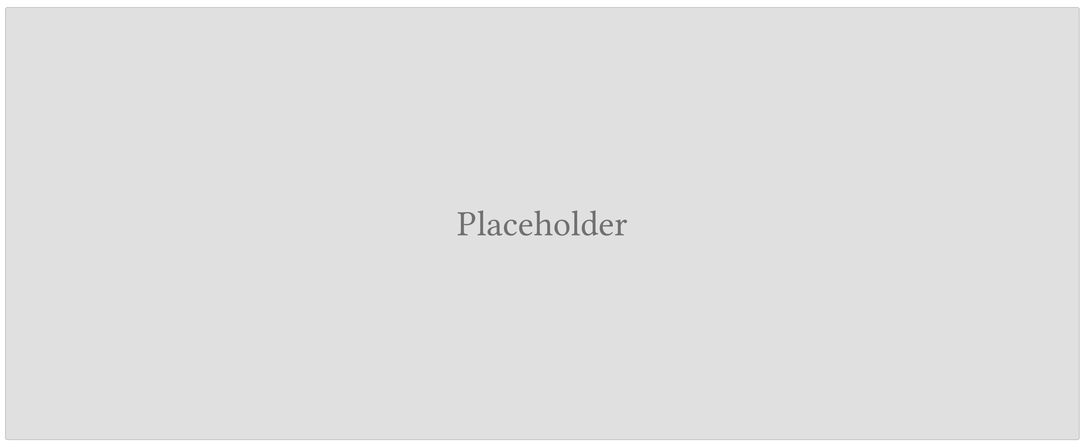

Figure 4:

Bibliography

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